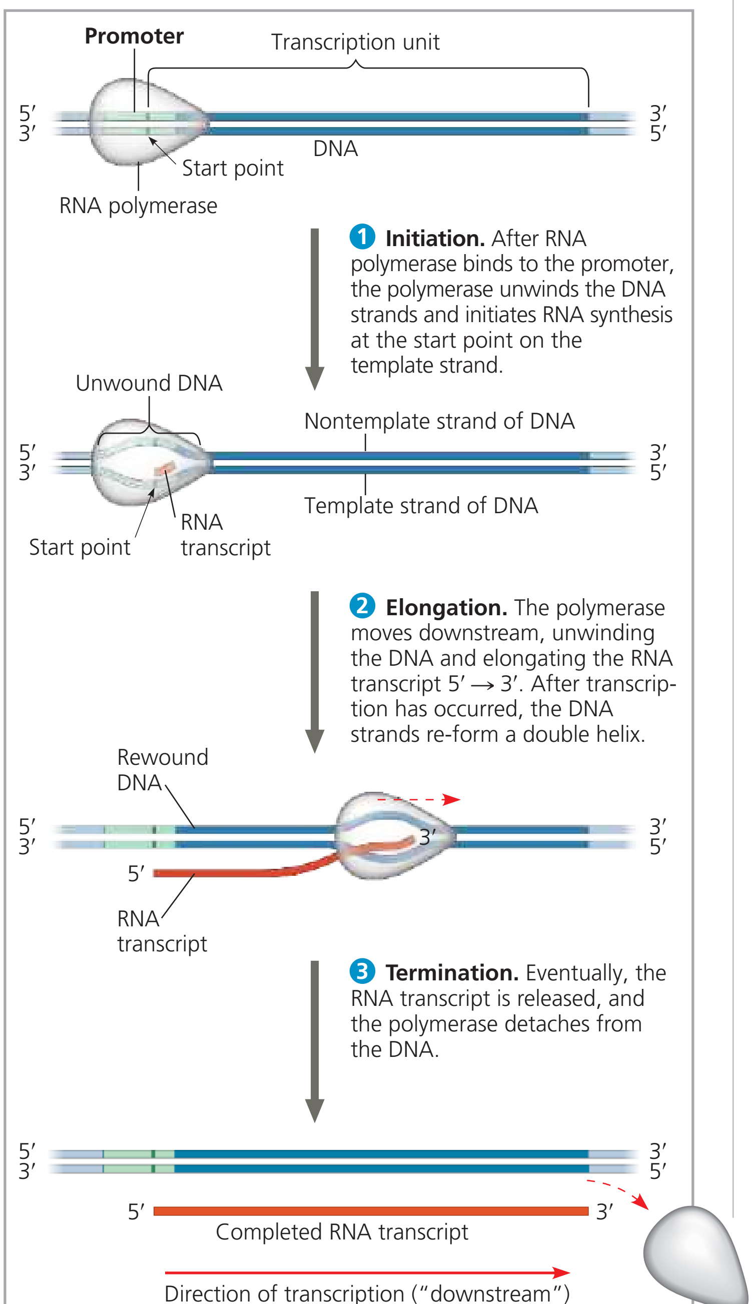
RNA Polymerase Binding and Initiation of Transcription

The promoter sequence of a gene includes within it the transcription **start point**—the nucleotide where RNA polymerase actually begins synthesizing the mRNA—and typically extends several dozen or so nucleotide pairs upstream from the start point (**Figure 17.9**). Based on interactions with proteins (transcription factors), RNA polymerase binds in a precise location and orientation on the promoter. This binding determines where transcription starts and the direction it will travel, thus which strand of DNA is used as the template.

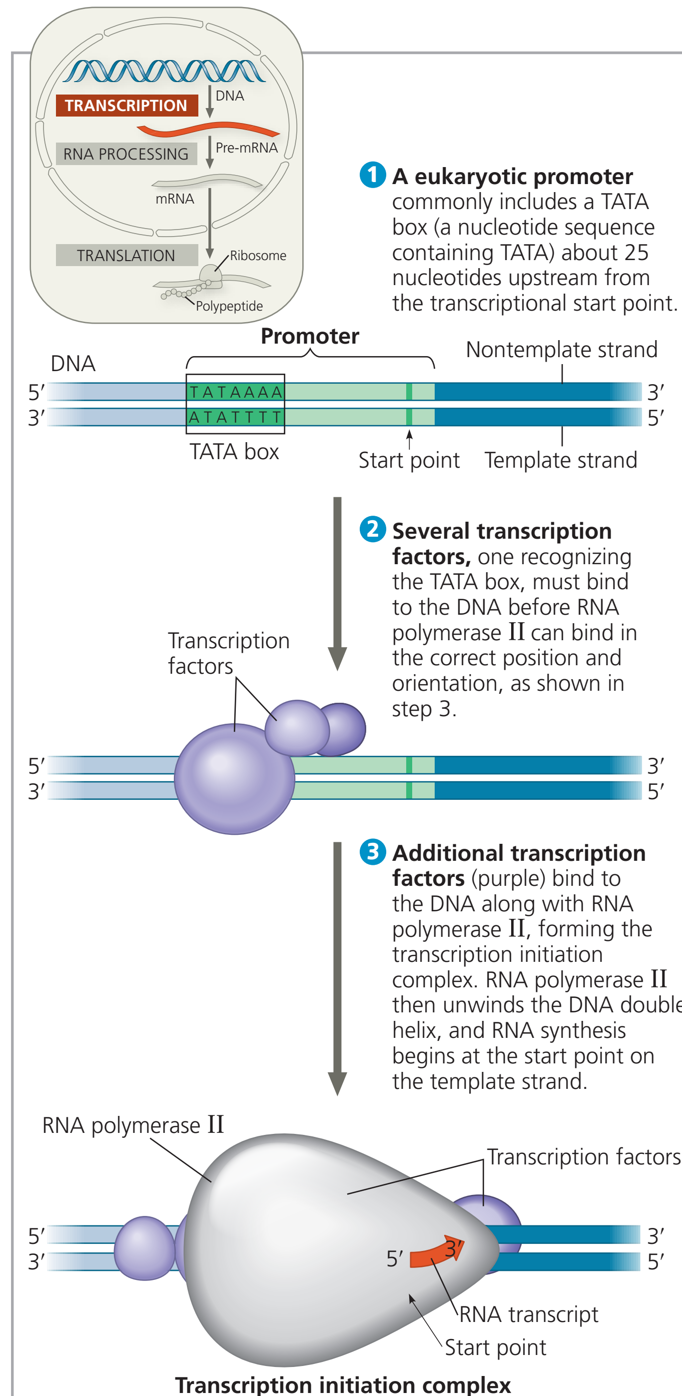
▼ Figure 17.8 The stages of transcription: initiation, elongation, and termination. This general depiction of transcription applies to both bacteria and eukaryotes, but the details of termination differ, as described in the text. Also, in a bacterium, the RNA transcript is immediately usable as mRNA; in a eukaryote, the RNA transcript must first undergo processing.



MAKE CONNECTIONS *Compare the use of a template strand during transcription and replication. See Figure 16.18.*

Mastering Biology **Animation: Overview of Transcription**
Animation: Overview of Transcription in Bacteria

▼ Figure 17.9 The initiation of transcription at a eukaryotic promoter. In eukaryotic cells, proteins called transcription factors mediate the initiation of transcription by RNA polymerase II.



? *Explain how the interaction of RNA polymerase with the promoter would differ if the figure showed transcription initiation for bacteria.*